Introducción a la secuenciación genómica de microorganismos de importancia en salud pública: Tecnología Illumina

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Marzo, 2026

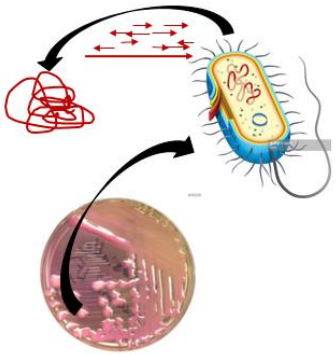
Bases del análisis de WGS

Para NGS se empieza con un genoma (libro completo), se descompone en millones de pedazos (sequence reads) y se trata de volver a construir utilizando algoritmos matemáticos.

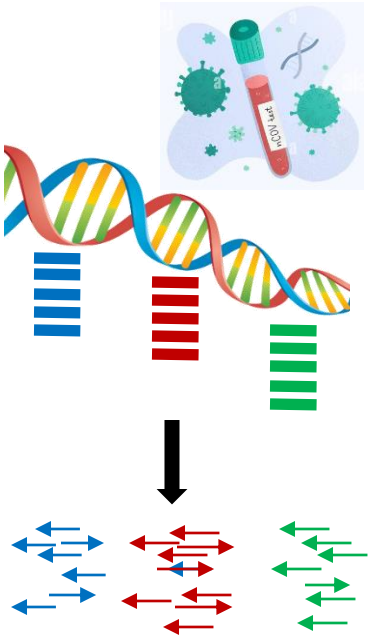
2016 Carleton, H. CDC

NGS Applications for Infectious Diseases

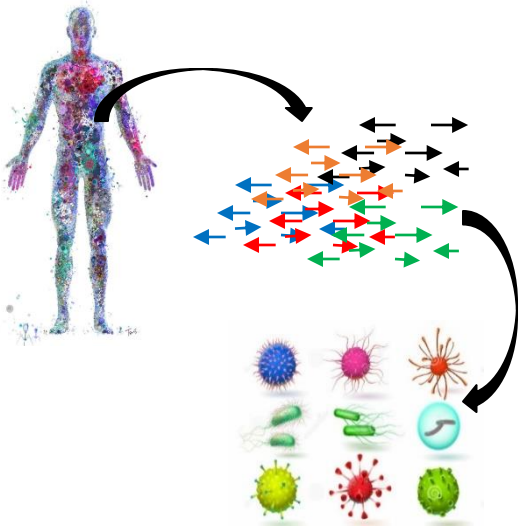
A. Whole Genome Sequencing



B. Targeted NGS (tNGS)

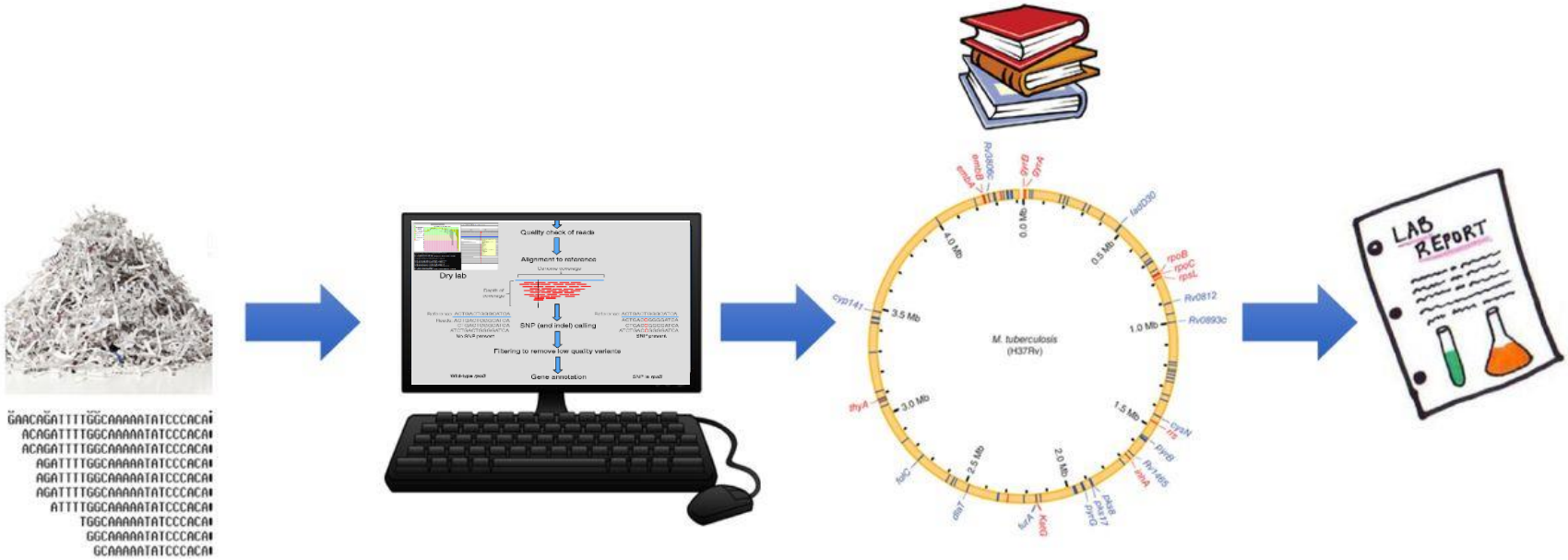


C. Metagenomic NGS (mNGS)



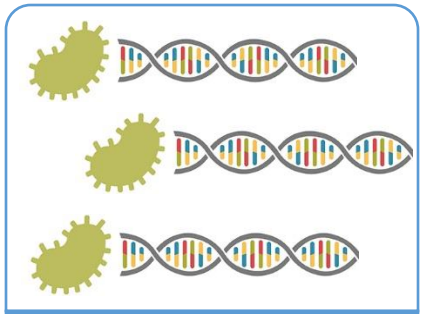
Estrategias para secuenciación de microorganismos para diagnóstico y seguimiento de enfermedades infecciosas

A. Whole Genome Sequencing



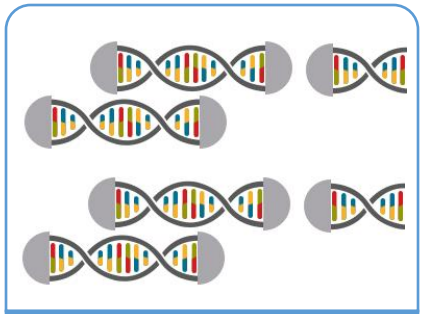
Escuyer, V. Use of Whole-Genome Sequencing for Detection of Antimicrobial Resistance: *Mycobacterium tuberculosis*, a Model Organism. Society for Clinical Laboratory Science Apr 2019, 32 (2) 85-92; **DOI:** 10.29074/ascls.2019001784

El proceso de secuenciación del genoma



Extracción del genoma

1

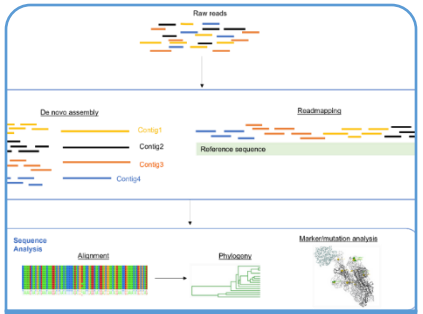


Preparación de bibliotecas

2

Secuenciación

3



Análisis bioinformático

4

Adaptado de: [What is Genomic Sequencing? | AMD | CDC]

DNA extraction and purification

Key quality indicators and quality control considerations:

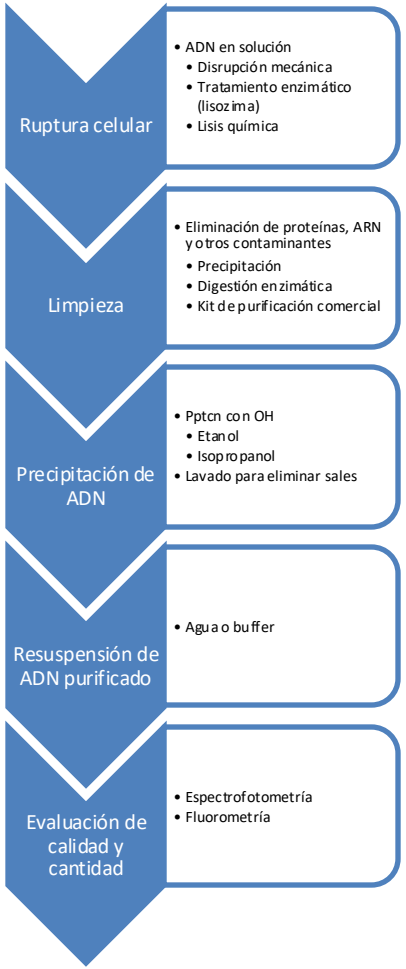
- Specimen/culture sample quality: adequate sample purity (with consideration for other organism DNA, human DNA or inhibitors in the sample)
- Specimen/sample quantity: Sufficient starting material (with consideration for targeted NGS or WGS application)
- DNA quality: Adequate DNA purity (spectrophotometer, $OD_{260/280}$ 1.8-2.0; $OD_{260/230}$ 2.0-2.2)
- DNA quantity: Sufficient starting material (fluorescent detection system, e.g. a Qubit® fluorometer or qPCR); Adequate volume
- DNA size and integrity (agarose gel electrophoresis or microfluidic instruments, including Bioanalyzer)

NOTES:

✓ DNA quantity and quality requirements are dependent on the sequencing application and platform. DNA Quantity/quality assessment must be always conducted after extraction step.

✓ DNA extraction procedures require adequate biosafety containment level when starting from direct specimens or cultured isolates, including appropriate safety equipment and work practices

Supporting information: *World Health Organization; 2018 (WHO/CDS/TB/2018.19)*



ADN intacto en alta concentración* y con baja cantidad de contaminantes:

- 260/280: 1,8-2,0 (valores bajos indican contaminación de proteína)
- 260/230: 2,0-2,2 (mide la pureza de los ácidos nucleicos y es un indicador de la contaminación por sales, fenol o carbohidratos)
- Cn para DNAPrep:
  - Min 4 ng/ul
  - Ideal > 10 ng/ul

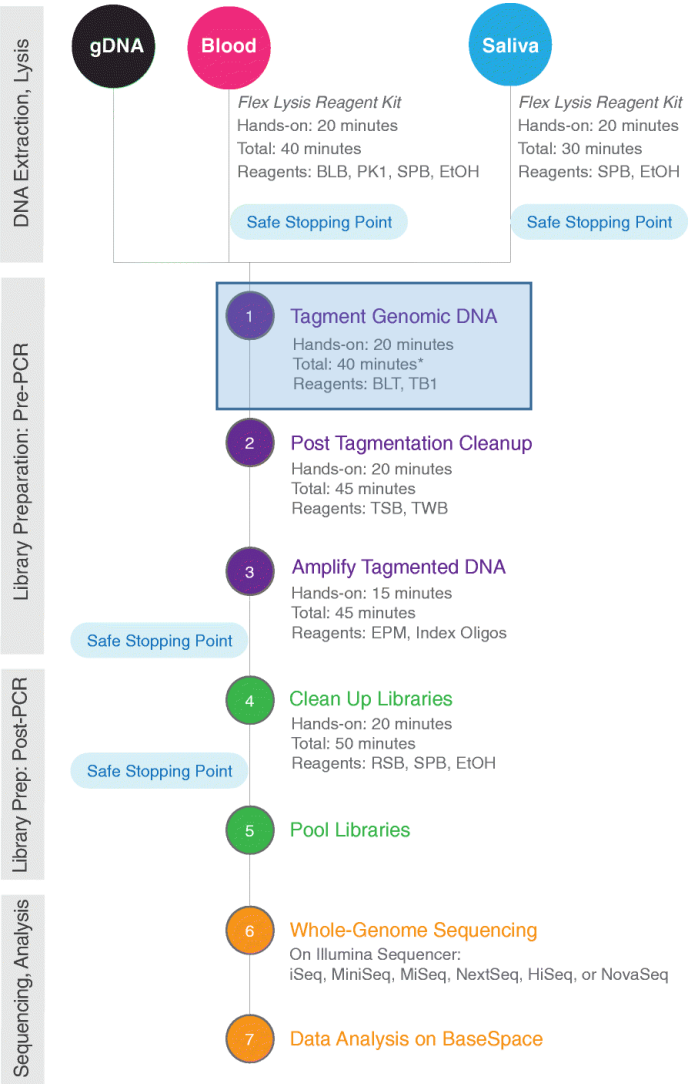
Illumina Bead-Linked Transposome Chemistry

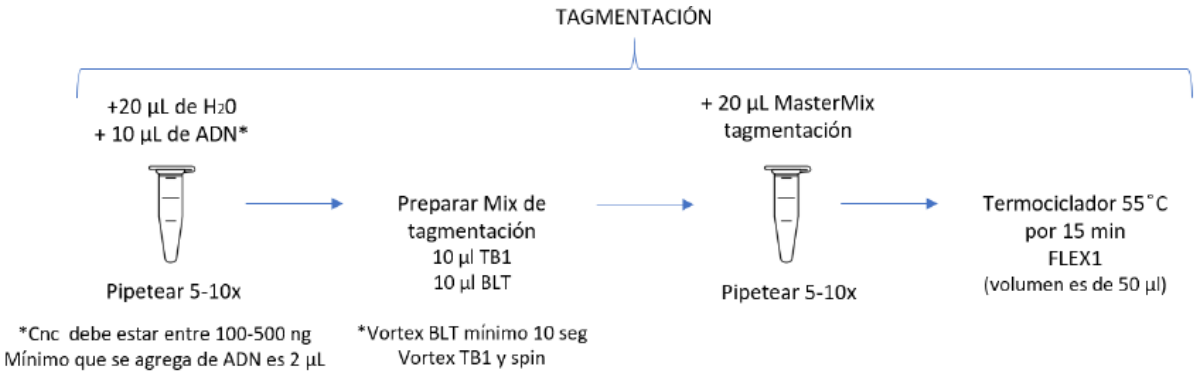
For clustering:
Libraries must have P5 and P7 binding regions on either end of a library

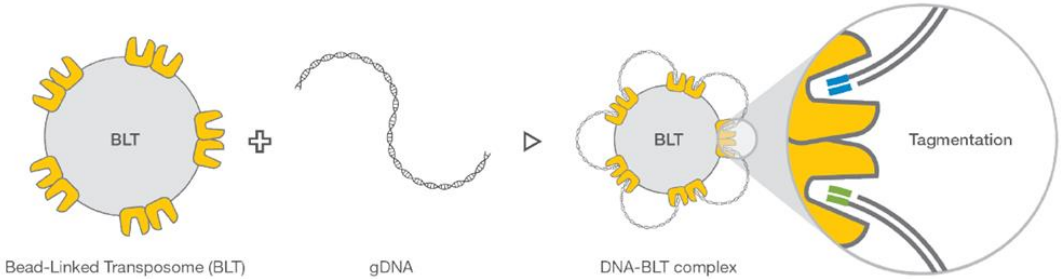
For sequencing:
Libraries must have sequencing primer binding regions

For mixing samples:
Libraries must have a unique index or barcodes sequence

Preparación de bibliotecas para secuenciación: Illumina DNAprep kit







Bruinsma, S., Burgess, J., Schlingman, D. *et al.* Bead-linked transposomes enable a normalization-free workflow for NGS library preparation. *BMC Genomics* **19,** 722 (2018). https://doi.org/10.1186/s12864-018-5096-9

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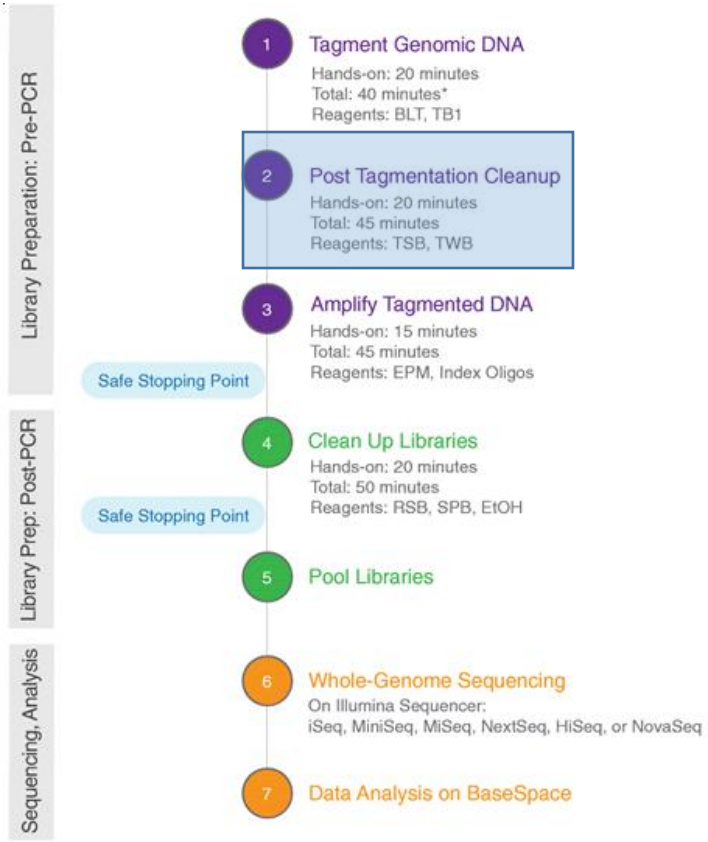

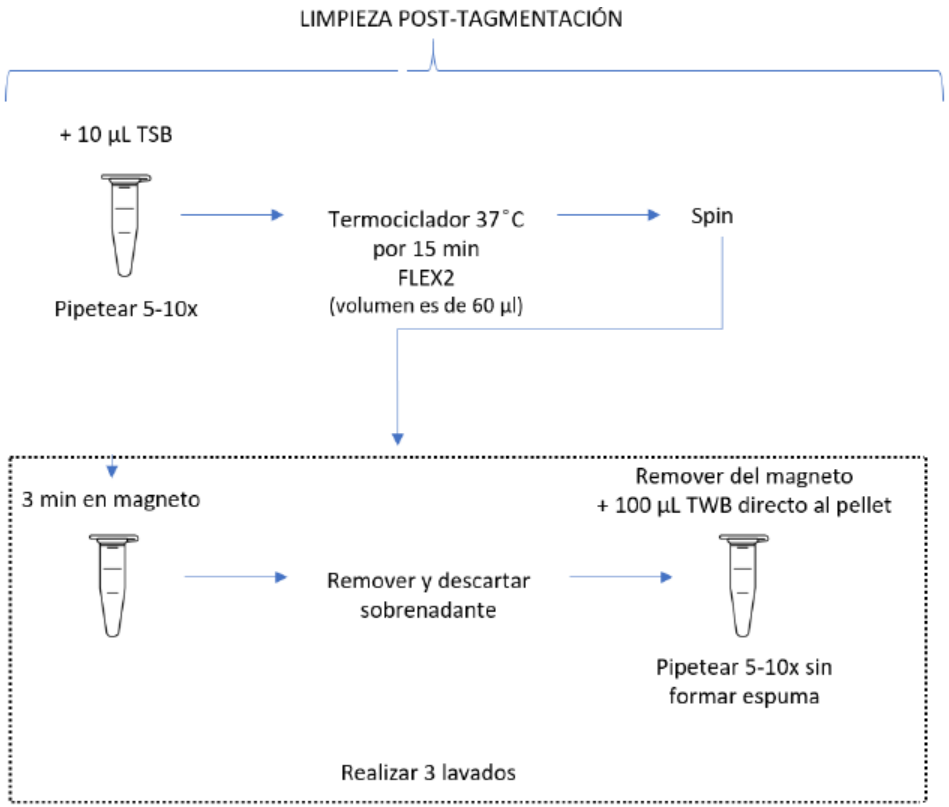

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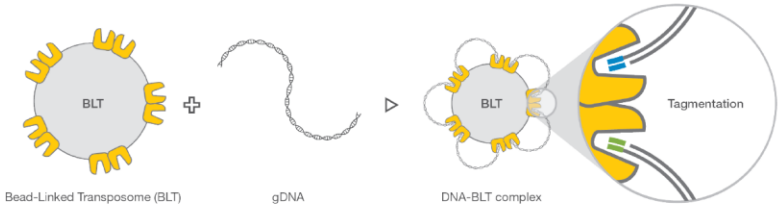

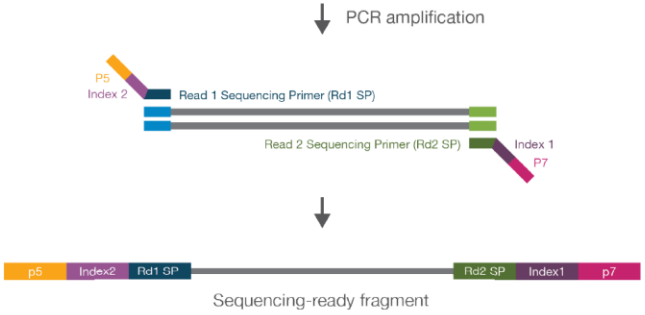

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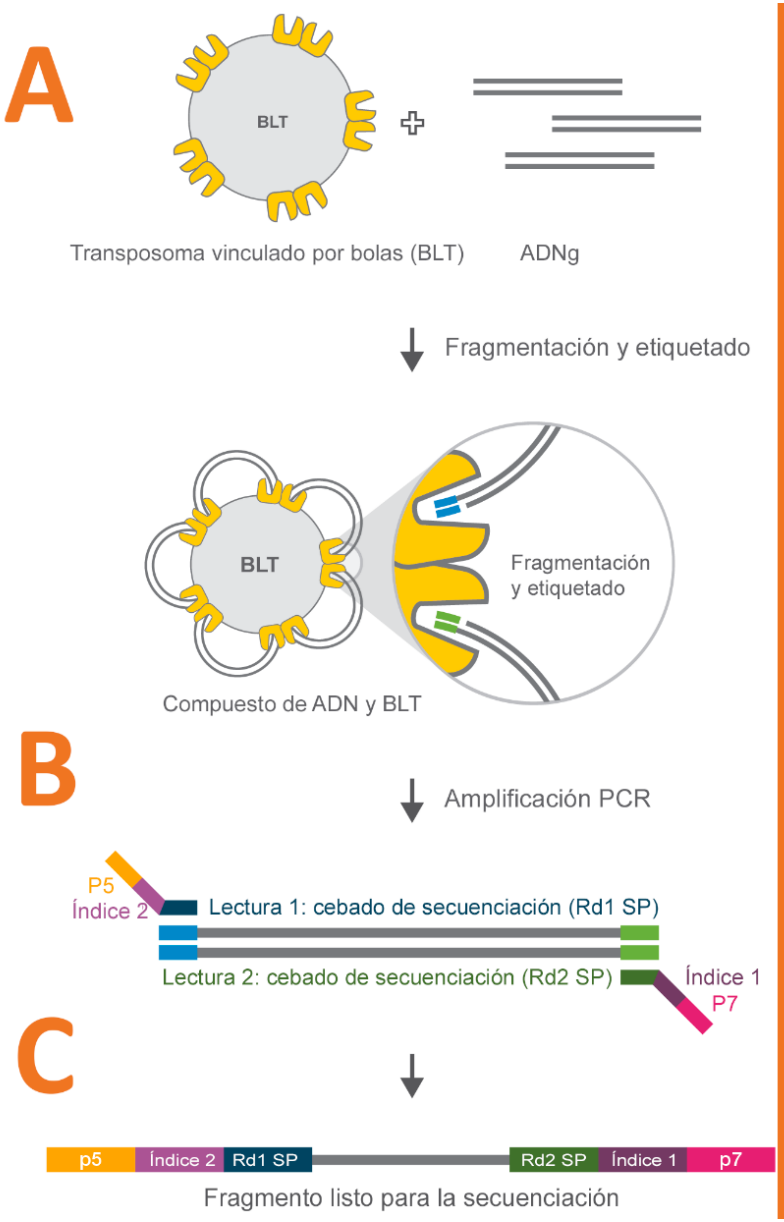
Procesos químicos de transposomas vinculados por bolas Nextera.



A

Los transposomas vinculados por bolas actúan como mediadores entre la fragmentación simultánea de ADNg y la adición de cebadores de secuenciación

B

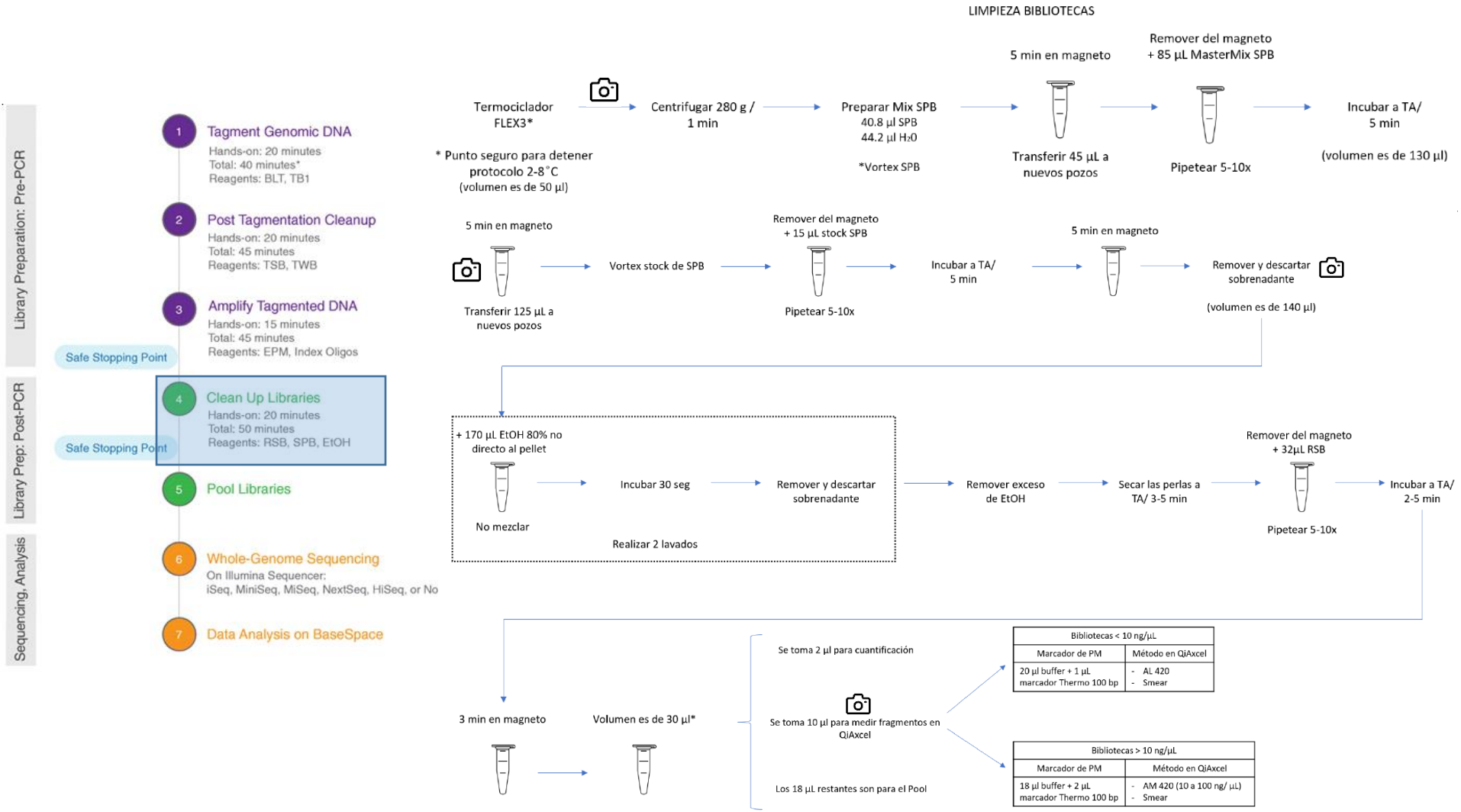
La PCR de ciclo reducido amplifica los fragmentos de ADN listos para la secuenciación y añade índices y adaptadores.

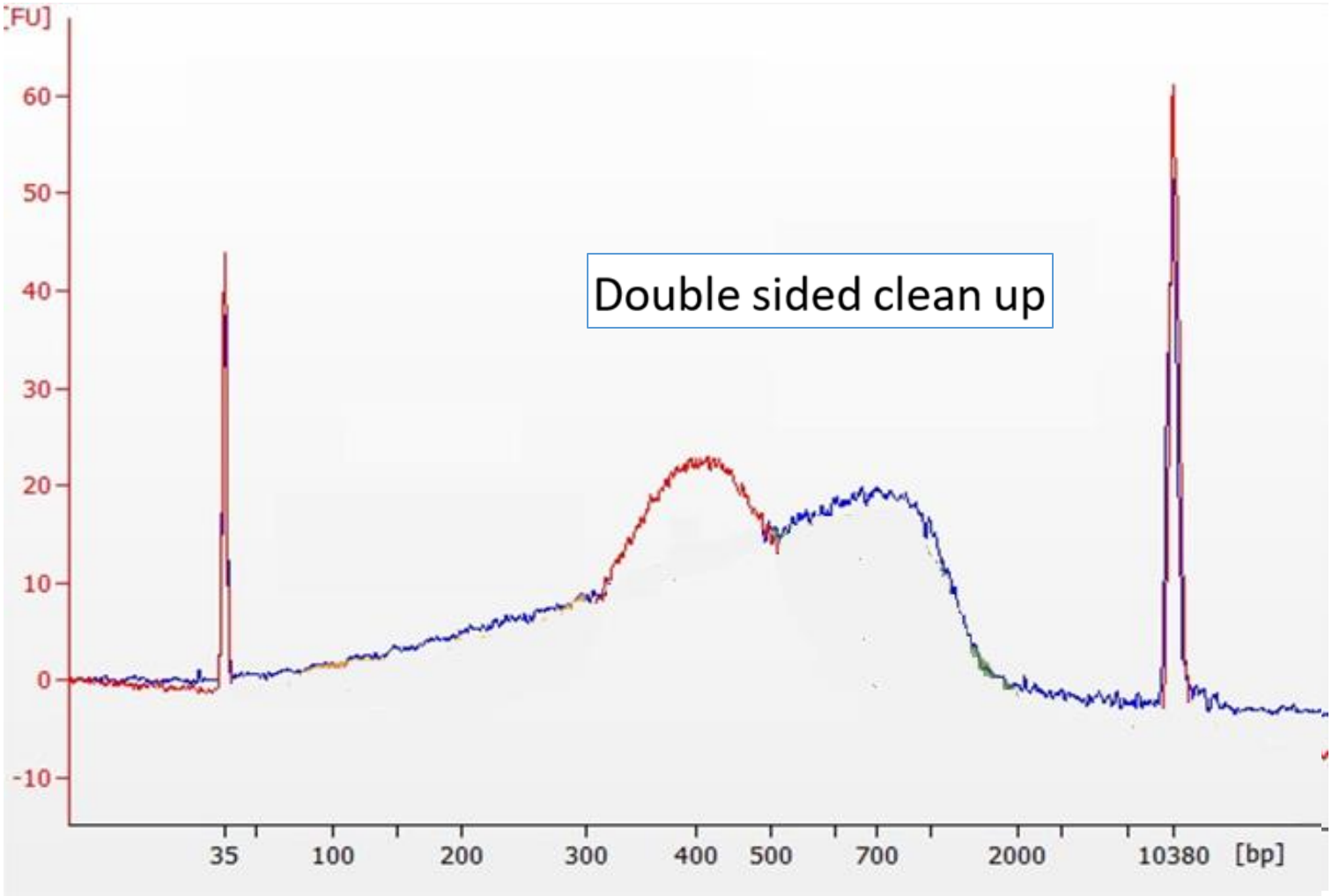
C

Los fragmentos listos para la secuenciación se lavan y se agrupan.

https://www.illumina.com/techniques/sequencing/ngs-library-prep/tagmentation.html

Preparación de bibliotecas para secuenciación: Illumina DNAprep kit





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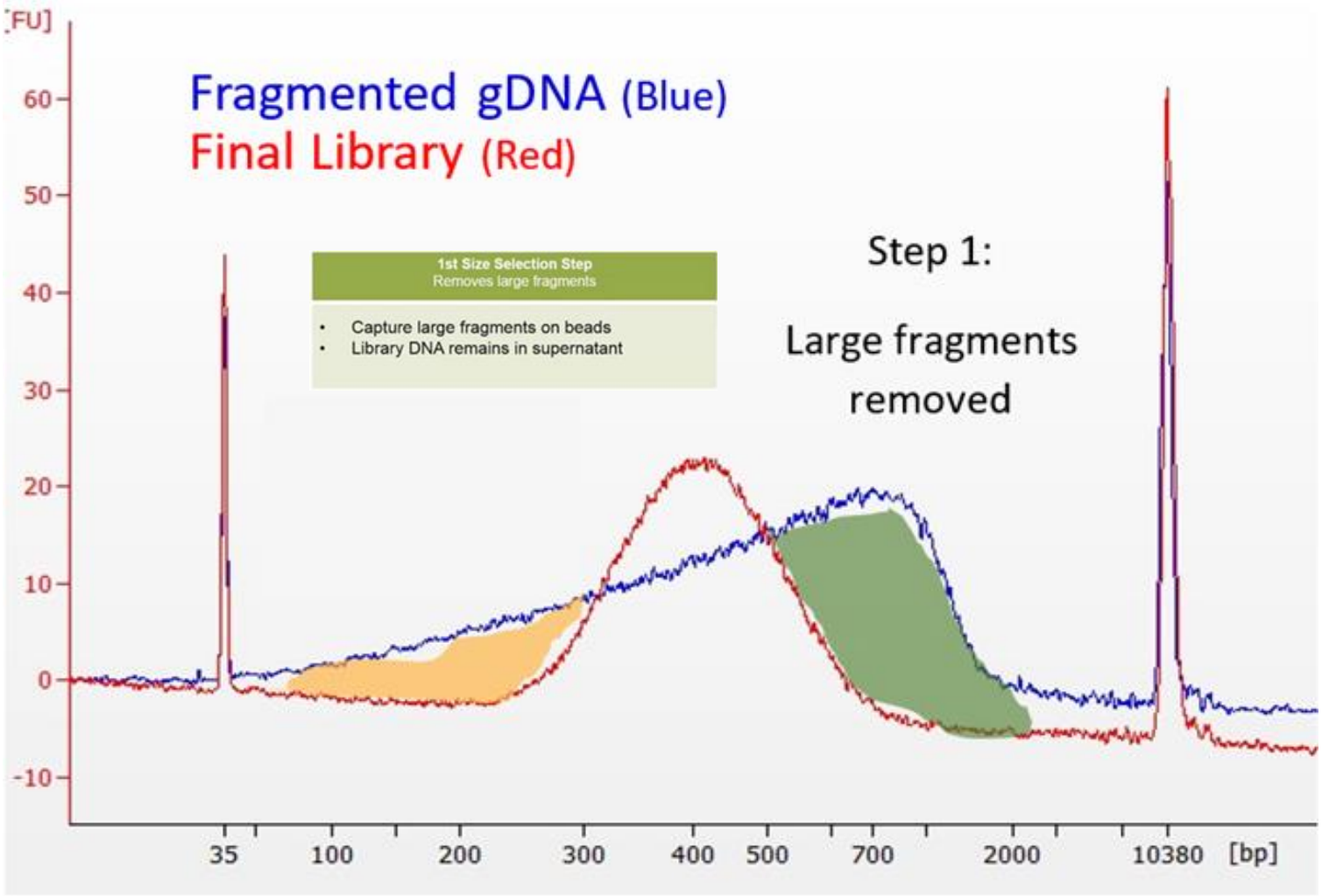

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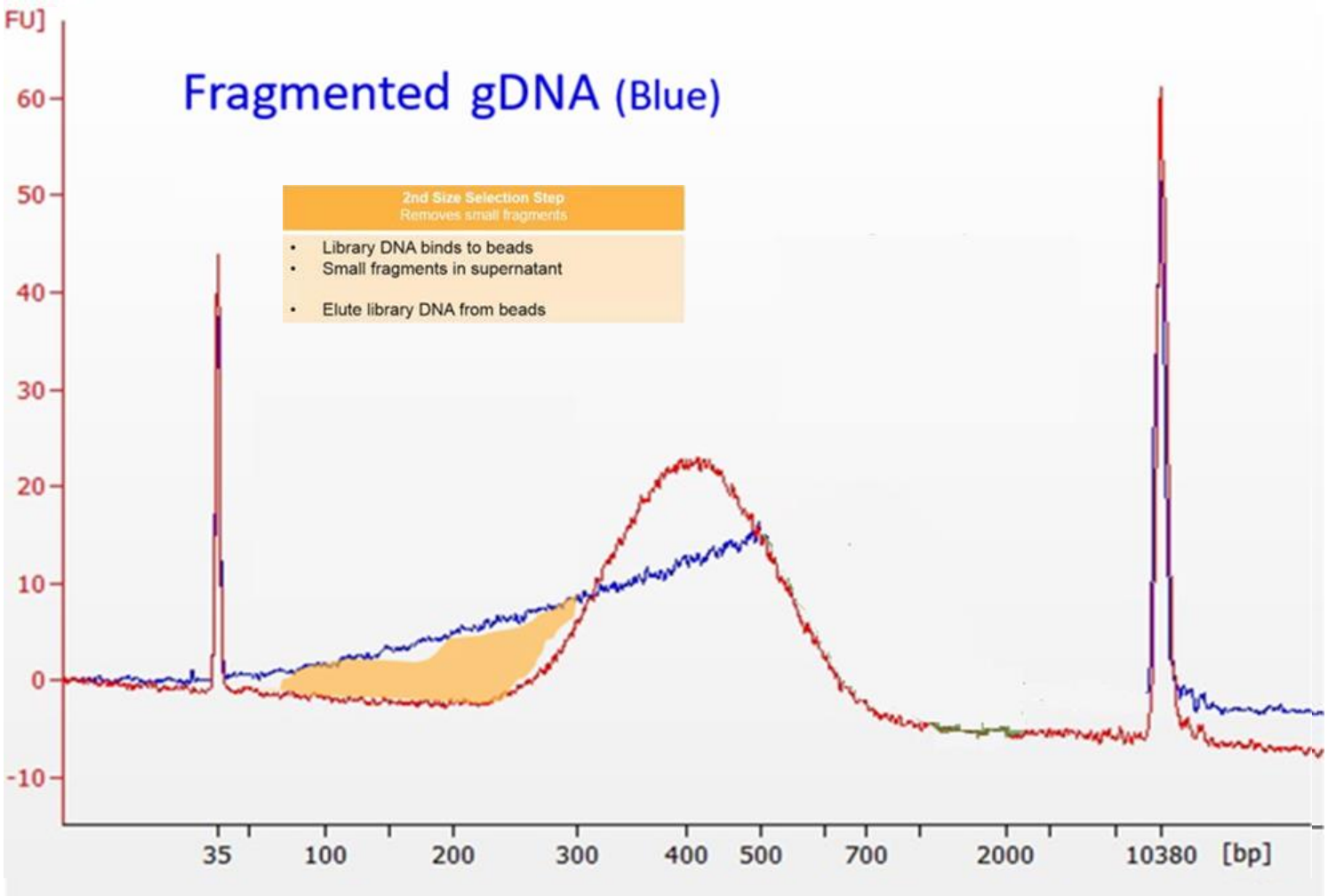

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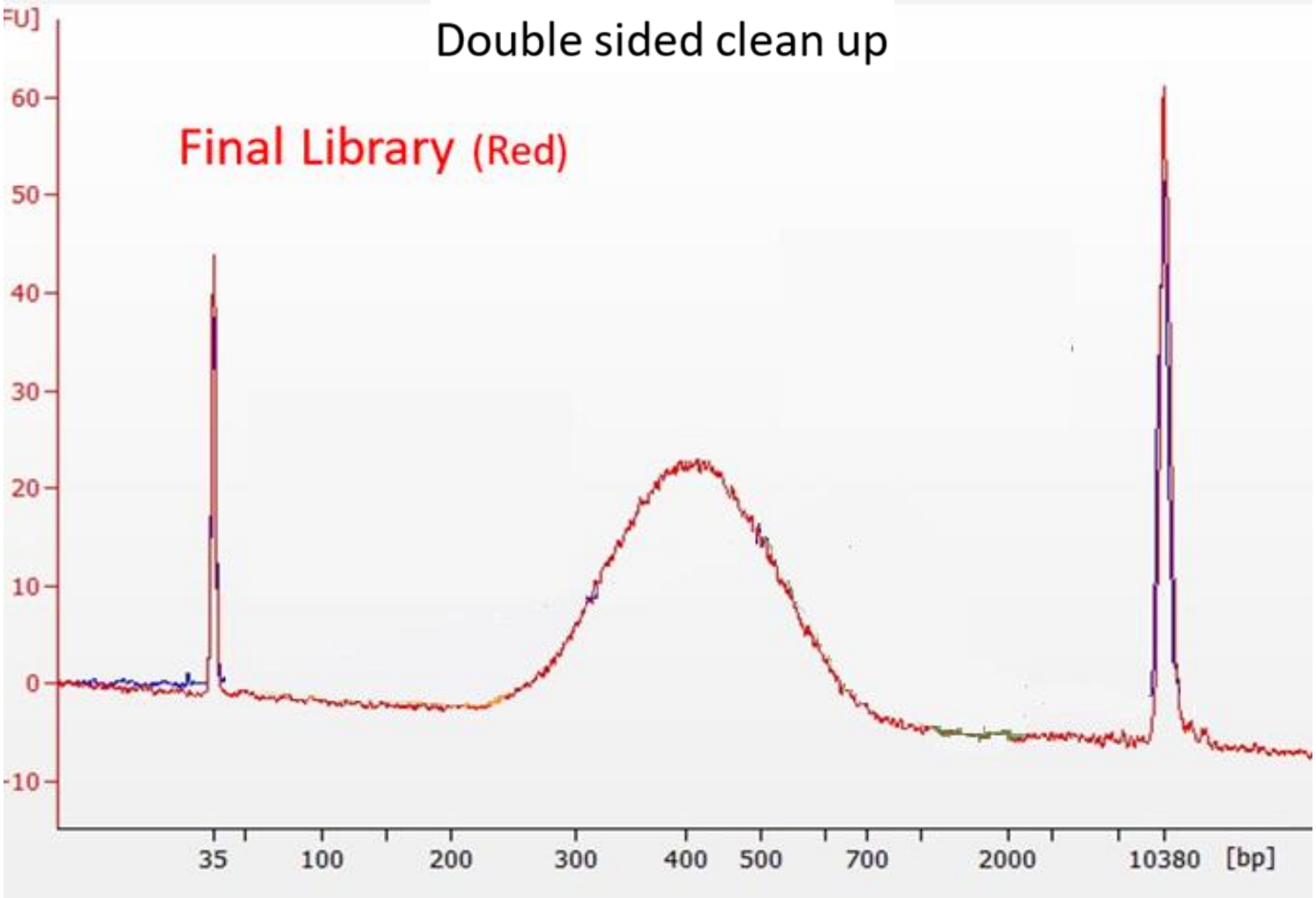

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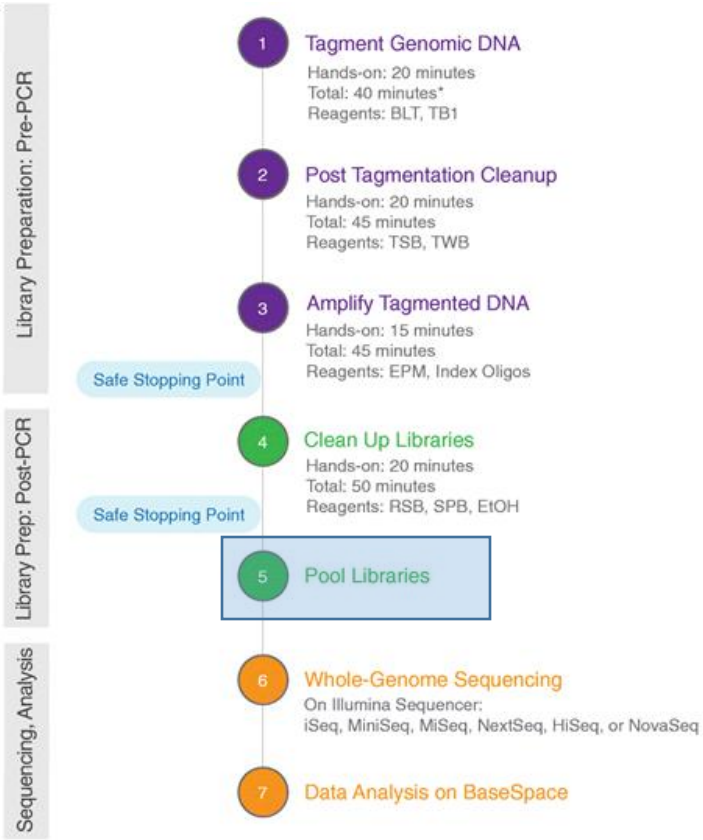

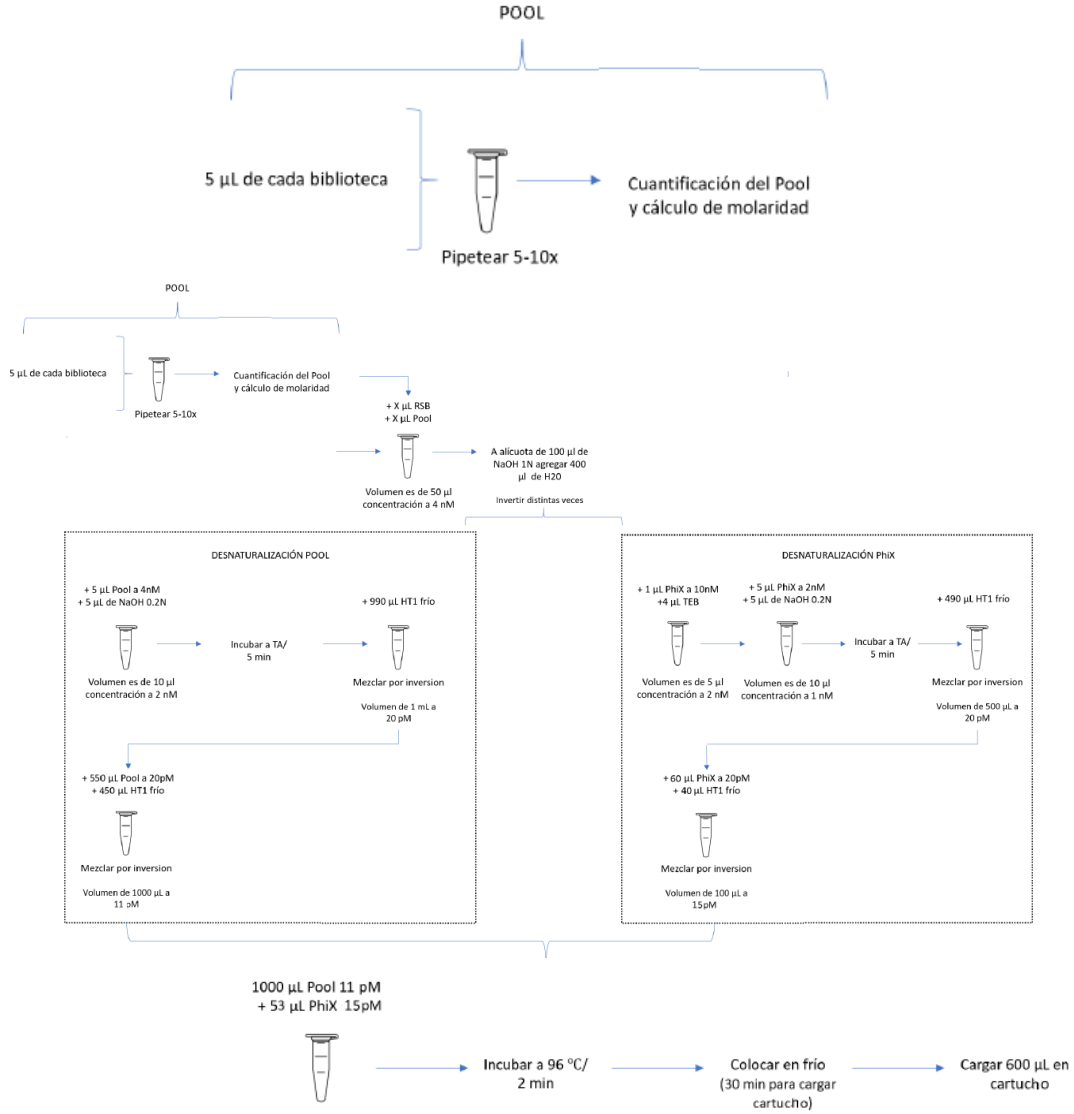

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Secuenciación por síntesis: Illumina

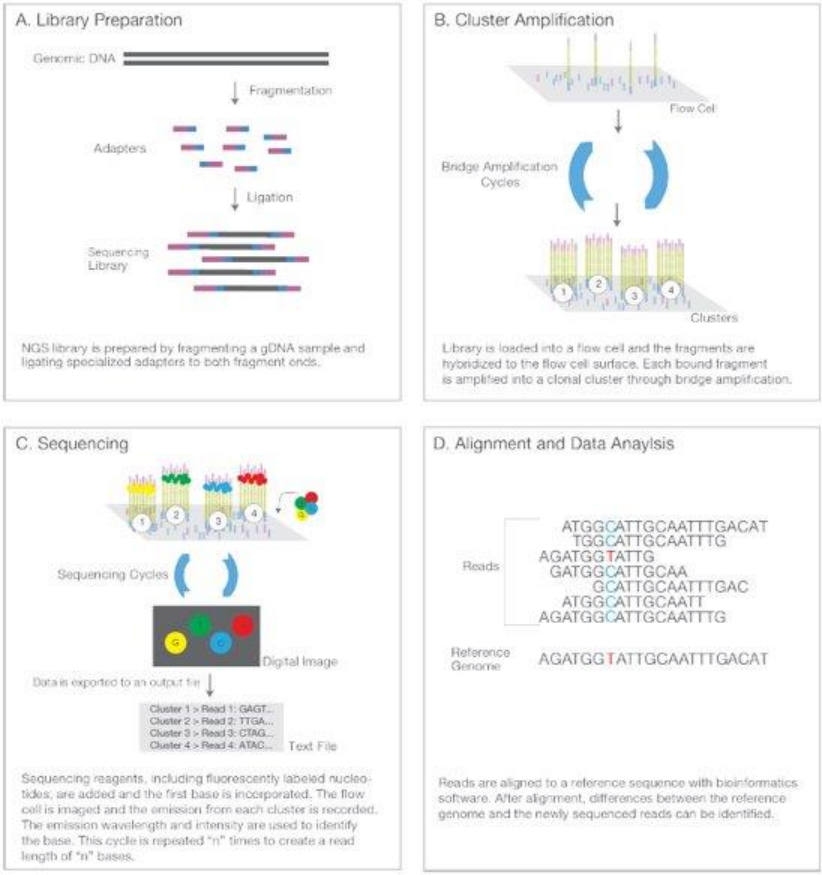
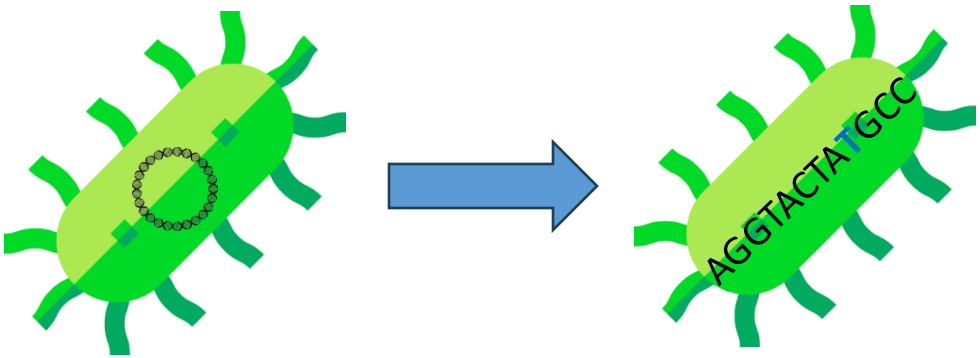


Figure 3: Next-Generation Sequencing Chemistry Overview—Illumina NGS includes four steps: (A) library preparation, (B) cluster generation, (C) sequencing, and (D) alignment and data analysis.

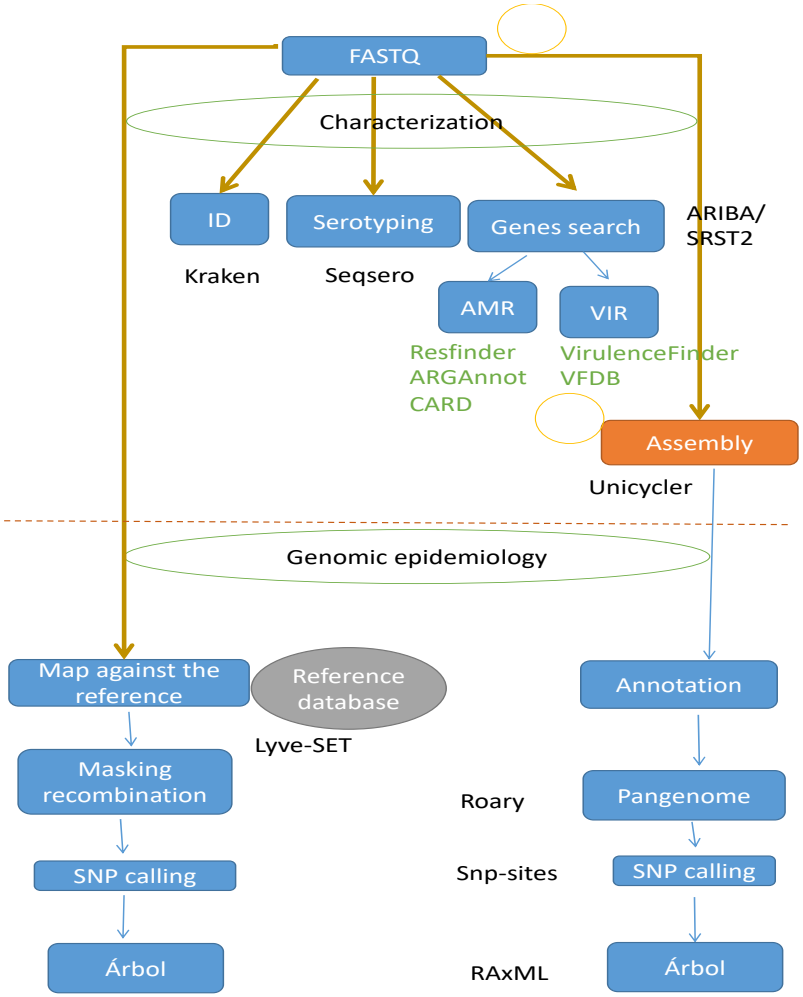
Bruinsma, S., Burgess, J., Schlingman, D. *et al.* Bead-linked transposomes enable a normalization-free workflow for NGS library preparation. *BMC Genomics* **19,** 722 (2018). https://doi.org/10.1186/s12864-018-5096-9

https://www.youtube.com/watch?v=fCd6B5HRaZ8&t=47s

¿Qué obtenemos y para qué nos sirve?

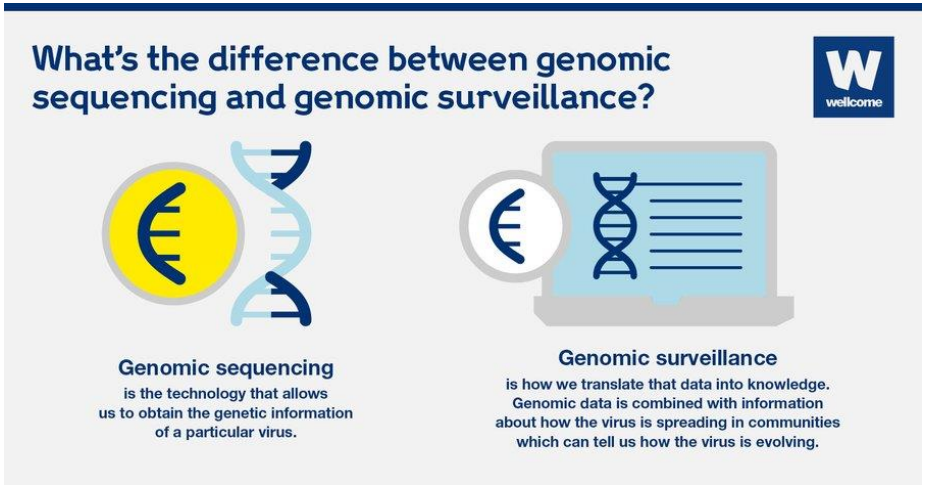


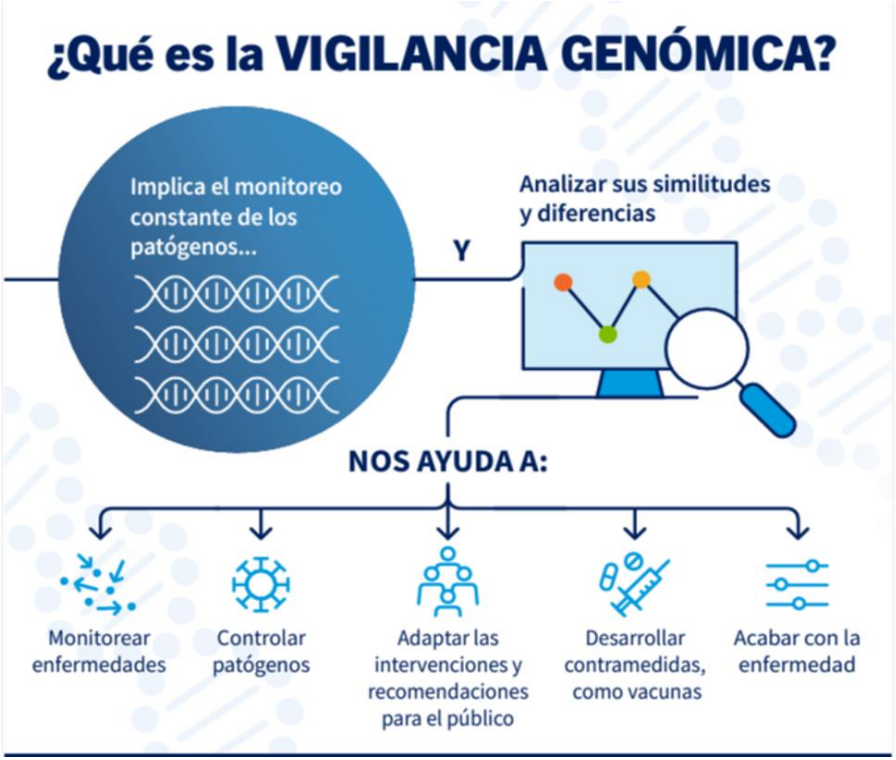
- Confirmar ID
- Tipificación y subtipificación
- Resistencia y virulencia
- Comparaciones genéticas



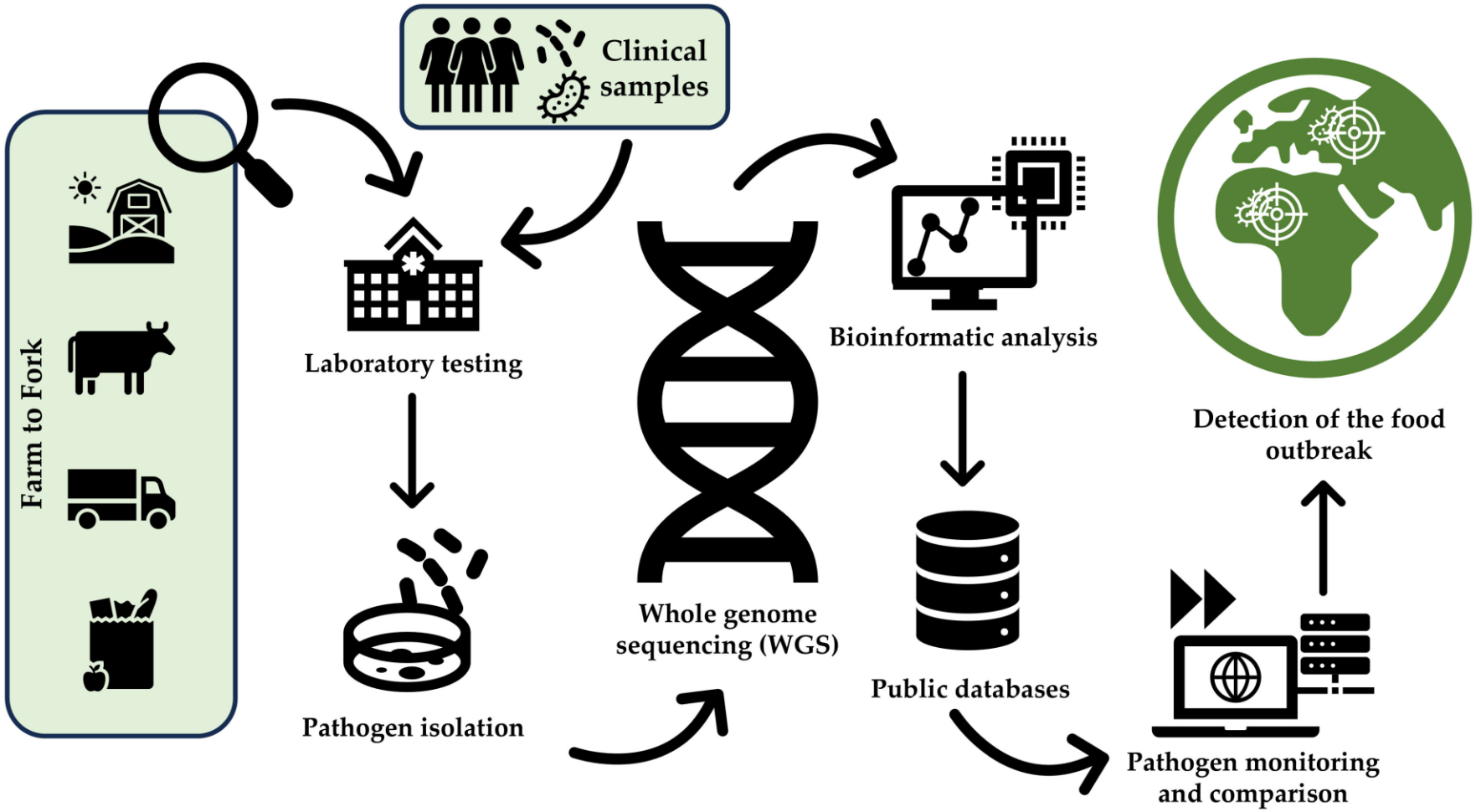
Vigilancia genómica

- Secuenciación genómica vrs Vigilancia genómica





Fundamentos de la vigilancia genómica aplicada a ETA



Gomes E, Araújo D, Nogueira T, Oliveira R, Silva S, Oliveira LVN, Azevedo NF, Almeida C and Castro J (2025) Advances in whole genome sequencing for foodborne pathogens: implications for clinical infectious disease surveillance and public health. Front. Cell. Infect. Microbiol. 15:1593219. doi: 10.3389/fcimb.2025.1593219

Agradecimientos

"La genómica no sustituye al trabajo de campo, lo potencia: nos da un mapa genético que, combinado con la epidemiología, permite actuar con más rapidez y precisión para proteger la salud pública."

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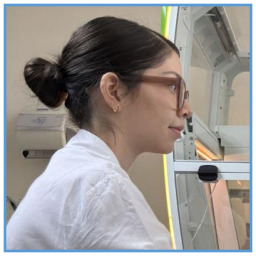
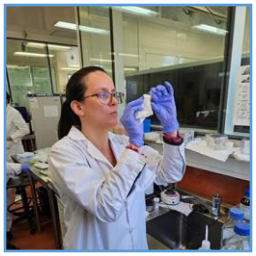
INSTITUTO CONMEMORATIVO GORGAS
DE ESTUDIOS DE LA SALUD (1928)

INCIENSA
Instituto Costarricense de Investigación y Enseñanza en Nutrición y Salud

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